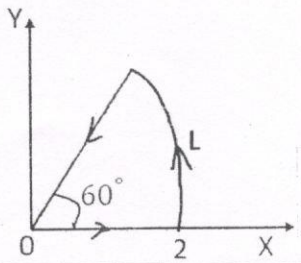
(Please write your Enrolment No. immediately) Enrolment No._____

MID TERM EXAMINATION
B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
IVth Semester, April, 2024

Paper Code: ECC 216 **Subject: Electromagnetic Field Theory**

Time :- $1\,^1/_2$ Hrs. **Max. Marks: 30**

Note: Attempt Q. No.1 Which is compulsory and any two more questions from remaining.

Q.No.	Question	Max. Marks	CO(s)
1 (a)	Calculate the circulation of $\vec{A} = \rho cos\emptyset \widehat{a_\rho} + z sin\emptyset \widehat{a_z}$ around the edge L of the wedge defined by $0 \le \rho \le 2, 0 \le \emptyset \le 60^0, z = 0$ and shown in Figure. (*use only Stokes's Theorem*) 	2.5	CO1
1 (b)	Explain physical significance of gradient, divergence and curl.	2.5	CO1
1 (c)	Find electric field at a point 'P' due to a point charge 'Q' located at origin. Use Gauss law only.	2.5	CO1
1 (d)	State Biot-Savart's law	2.5	CO2
2 (a)	Given point *P* (-2,6,3) and vector $\vec{A} = y\widehat{a_x} + (x+z)a_y$, express *P* and $\vec{A}$ in cylindrical coordinates system.	4	CO1
2 (b)	State and prove Divergence and Stokes's Theorem	6	CO1
3 (a)	Explain Gauss's Law and derive first law of Maxwell's equation.	5	CO1
3 (b)	Given the potential $V = \frac{10}{r^2} sin\theta cos\emptyset$. Find the electric flux density D at $(2, \frac{\pi}{2}, 0)$.	5	CO1
4 (a)	Explain the magnetic boundary condition between two different dielectric media.	5	CO2
4 (b)	State Ampere's Circuit law and derive third Maxwell's equation	5	CO2